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**NMAM INSTITUTE OF TECHNOLOGY, NITTE**  
**Off-Campus Centre of Nitte (Deemed to be University)**  
**Second Semester B.Tech (CBCS) Degree Examinations**

May 2025

**BT1002-1 – CELL BIOLOGY**  
(For BT)

Duration: 3 Hours

Max. Marks: 100

**Note:**

**Part – A: Multiple Choice Questions:** Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

**Part – B: Descriptive Answer Questions:** Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

**PART - A: MULTIPLE CHOICE QUESTIONS****20 Marks**

1. The Polymerase Chain Reaction, a method used to amplify a copy of a segment of a DNA was invented by \_\_\_\_\_  
A) Frederick Sanger  
B) Rosalind Franklin  
C) Kary Mullis  
D) Karl Rekey
2. Which of the following is not a part of the Cell Theory?  
A) All living things are composed of cells  
B) Cells are the basic units of structure and function  
C) Cells can produce energy  
D) Cells come from pre-existing cells
3. Which of the following is responsible for the movement of cilia and flagella?  
A) Centrioles  
B) Mitochondria  
C) Ribosomes  
D) Golgi apparatus
4. Which of the following is the site of oxidative phosphorylation in mitochondria?  
A) Matrix  
B) Inner membrane  
C) Outer membrane  
D) Intermembrane space
5. Which of the following is not a type of lipid found in the cell membrane?  
A) Phospholipids  
B) Cholesterol  
C) Triglycerides  
D) Glycolipids
6. Which of the following is a disorder caused by dysfunction of lysosomes?  
A) Alzheimer's disease  
B) Cystic fibrosis  
C) Tay-Sachs disease  
D) Sickle cell anemia
7. During which phase of mitosis do the chromosomes separate and move to opposite poles of the cell?  
A) Anaphase  
B) Metaphase  
C) Prophase  
D) Telophase
8. In electron microscopy, which of the following can be observed?  
A) Glass lens  
B) Gap junctions  
C) Plastic lens  
D) Fiber lens
9. Which type of junction provides a strong physical connection between adjacent cells?  
A) Tight junctions  
B) Gap junctions  
C) Adherens junctions  
D) Desmosomes
10. Which type of protein is responsible for providing structural support to the extracellular matrix?  
A) Collagen  
B) Actin  
C) Myosin  
D) Keratin

11. Which type of cell signaling involves the release of signaling molecules into the bloodstream to act on distant target cells?  
 A) Autocrine signaling B) Paracrine signaling  
 C) Endocrine signaling D) Synaptic signaling
12. What is the structure of a G protein-linked receptor?  
 A) It has a single transmembrane domain and an intracellular catalytic domain. B) It has seven transmembrane domains and an extracellular ligand-binding domain.  
 C) It has a single transmembrane domain and an extracellular ligand-binding domain. D) It has seven transmembrane domains and an intracellular G protein-binding domain.
13. Which of the following is a common second messenger in animal cell signaling?  
 A) Calcium ions B) DNA  
 C) Nitric oxide D) Ethylene
14. Which of the following is an example of a microbial diagnostic test that uses PCR technology?  
 A) Blood test for cholesterol levels B) Urine test for glucose levels  
 C) Stool test for bacterial infections D) All of the these
15. Which of the following is a characteristic of stem cells?  
 A) They are genetically identical to the donor organism B) They can divide and differentiate into multiple cell types  
 C) They can be derived only from embryos D) They can differentiate into only one cell type
16. George Grey had successfully made the first continuous cell line culture. The cell line was derived from the \_\_\_\_\_ of Henrietta Lacks.  
 A) Cervical cancer cells B) Ovary cancer cells  
 C) Breast cancer cells D) Lung cancer cells
17. The field where living systems are analyzed within the context of other systems is known as  
 A) Systems biology B) Bioinformatics  
 C) Molecular biology D) Structural biology
18. 21 trisomy chromosome is the condition of -----  
 A) Extra ordinary B) Normal chromosome  
 C) Syphilis D) Down syndrome
19. Microtubule hollow tube wall consists of ----- columns of tubulin molecules  
 A) 25 B) 20  
 C) 13 D) 15
20. \_\_\_\_\_ expressed in nervous system are responsible for the specificity of neuronal connections  
 A) E- Cadherins B) N - Cadherins  
 C) Selectin D) Integrin

### **PART - B: DESCRIPTIVE ANSWER QUESTIONS**

| Unit – I |                                                                                                                                                                          | Marks | BT* | CO* | PO*  |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|------|
| 1.       | a) Reflect on the historical development of the Cell Theory, and how did they contribute to our current understanding of cells and their functions?                      | 08    | L*1 | 1   | 1,12 |
|          | b) Examine the role of organelles in contributing to cell diversity. Provide examples of organelles that are specific to certain cell types and explain their functions. | 08    | L1  | 1   | 1,12 |
| 2.       | a) Explain the significance of the cytoplasmic matrix in cellular metabolism. How does it support the various biochemical reactions that take place within a cell?       | 08    | L1  | 1   | 1,12 |
|          | b) What are the key events that occur during Meiosis-I? How does meiosis contribute to genetic variation?                                                                | 08    | L1  | 2   | 4,12 |

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|-------------------------------------------------------------------------------------------------------------------------------------------------|-------|----|---|------|
| 3. a) Comment on basic principle of electron microscopy with the help of flow diagram.                                                          | 08    | L1 | 2 | 4,12 |
| b) Give a contrast note on: i) Live cell imaging and fixed cell imaging, ii) Transmission Electron Microscopy and Scanning Electron Microscopy. | 04+04 | L1 | 2 | 4,12 |

**Unit – II**

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|-------------------------------------------------------------------------------------------------------------------------|----|----|---|------|
| 4. a) Describe the three main types of elements that make up the cytoskeleton.                                          | 08 | L1 | 3 | 1,12 |
| b) Discuss about general principles of cell signaling process and add a note on different types of signaling molecules. | 08 | L1 | 4 | 1,12 |
| 5. a) Provide a brief overview on ECM with its components.                                                              | 08 | L1 | 3 | 1,12 |
| b) Cells receive and process a diverse set of chemical signals and sensory stimuli. Narrate.                            | 08 | L1 | 4 | 1,12 |
| 6. a) Comment on occluding and anchoring junctions with one example each.                                               | 08 | L1 | 3 | 1,12 |
| b) Narrate the differentiating feature on Microtubule and Actin filament.                                               | 08 | L1 | 3 | 1,12 |

**Unit – III**

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|-----------------------------------------------------------------------------------------------------------------|----|----|---|------|
| 7. a) How microbial cell play a role in developing product through fermentation? Explain with suitable example. | 08 | L1 | 5 | 1,12 |
| b) Comment on microbial biosensors for water and soil analysis.                                                 | 08 | L1 | 5 | 1,12 |
| 8. a) Discuss about microbial biomonitoring of environmental pollutants through biosensors.                     | 08 | L1 | 5 | 1,12 |
| b) Write the pros and cons of cellular modifications using modern tools.                                        | 08 | L1 | 5 | 1,12 |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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